HOTEL BOOKINGS CANCELLATION ANALYSIS

DATA-DRIVEN INSIGHTS & REVENUE OPTIMIZATION STRATEGIES

- **Prepared For:** Hospitality Operations & Revenue Management Teams
- **Analysis Basis:** 119,390 Booking Records (2015-2017)



1. Executive Summary

This report provides a data-driven analysis of a hotel bookings dataset to address a high **overall cancellation rate of 37.13%**, which severely impacts revenue and inventory. By analyzing hotel types, seasonal trends, average daily rates (ADR), geographical distributions, and market segments, this report identifies key drivers of booking churn. The insights offer actionable strategies to reduce cancellation risks and optimize room occupancy.

2. Problem Statement and Objectives

Problem Statement

The hotel industry faces substantial revenue leakage due to booking volatility. High cancellation rates complicate room inventory control, result in perishable rooms expiring unbooked, disrupt staffing, and distort demand-forecasting models. To maximize profitability, hotels must identify the drivers behind who cancels, when they cancel, and how pricing impacts customer behavior.

Objectives

- **Analyze Churn Dynamics:** Compare booking fulfillment and cancellation ratios between City and Resort Hotels.
- **Identify Seasonal Volatility:** Isolate monthly and seasonal trends in booking volumes and customer cancellation behaviour.

- **Evaluate Pricing Impact:** Examine the correlation between Average Daily Rate (ADR) and cancellation probabilities.
- **Pinpoint Geographical Hotspots:** Identify top countries of origin contributing to cancellations.
- **Formulate Strategic Interventions:** Provide data-backed recommendations on overbooking, deposits, and target marketing.

3. Dataset Description

The dataset contains 119,390 rows and 32 columns tracking booking performance across City and Resort Hotels from July 2015 to August 2017. Key metrics include reservation status, booking lead time, arrival dates, length of stay, guest demographics, market segments, distribution channels, deposit types, agent/company IDs, and Average Daily Rate (ADR).

4. Tools and Technologies

- **Python:** Core programming language used for data manipulation and statistical calculations.
- **Pandas:** Used for data profiling, filtering, row/column cleansing, and aggregations.
- **Matplotlib & Seaborn:** Utilized to engineer diagnostic plots (box plots, count plots, and time-series line charts).

5. Data Cleaning and Preprocessing

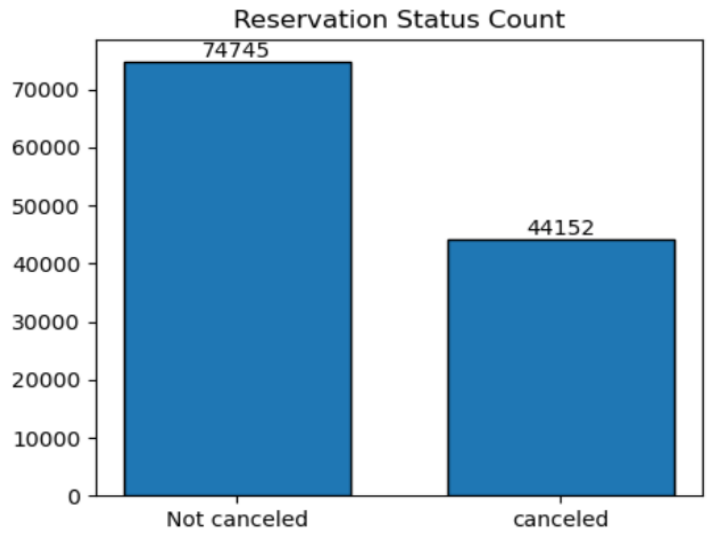
To secure data integrity, the dataset was processed through a strict cleaning pipeline:

- **Irrelevant Feature Removal:** The columns agent and company contained excessive missing data (16,340 and 112,593 null values) and were entirely dropped to eliminate systemic bias.
- **Handling Missing Values:** Rows containing null elements in country (488 records) and children (4 records) were removed, yielding a clean dataset of **118,898 valid records**.
- **Data Type Synchronization:** The reservation_status_date column was successfully converted from a text object into a synchronized datetime object.
- **Outlier Mitigation:** An initial audit of the Average Daily Rate (adr) highlighted a severe extreme outlier at 5,400.00. This single row was removed using a threshold restriction (adr < 5000), bringing the operating maximum ADR to a valid 510.00.

6. Reservation Status Analysis

The baseline distribution of the dataset highlights a major structural vulnerability in the booking pipeline:

- **Fulfilled Reservations (Check-Outs):** 62.87% (74,745 bookings) concluded with guests successfully completing their stays.
- **Canceled Bookings (Cancellations & No-Shows): 37.13%** (44,152 bookings) resulted in churn.

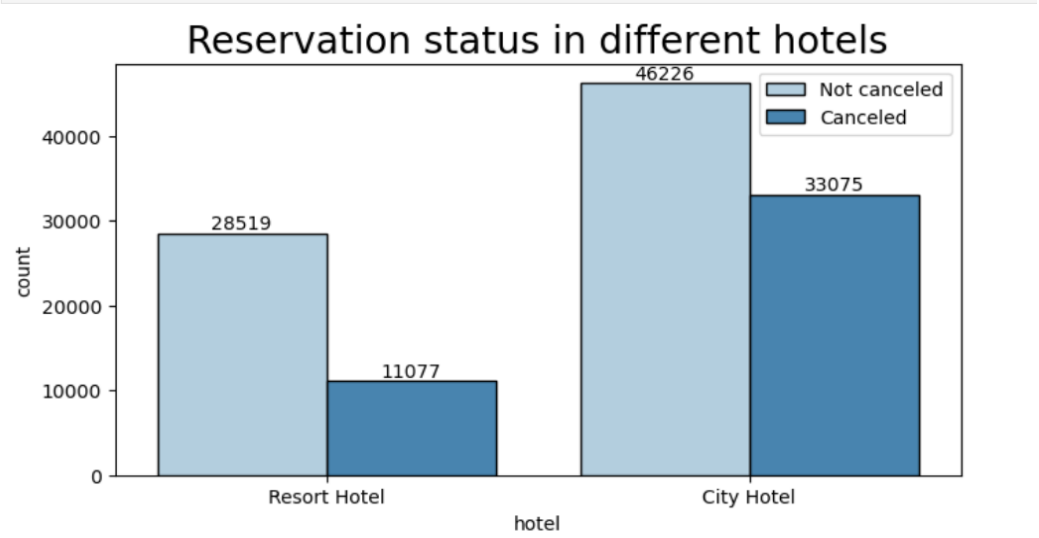


7. Hotel-wise Cancellation Analysis

Segmenting reservation performance by hotel type reveals that booking volatility is heavily concentrated within urban environments:

- **City Hotel:** 79,302 bookings | 46,226 check-outs | 33,076 cancellations (**41.71% cancellation rate**)
- **Resort Hotel:** 39,595 bookings | 28,518 check-outs | 11,077 cancellations (**27.97% cancellation rate**)

City Hotels are highly volatile. Nearly 42 out of every 100 city reservations turn into cancellations due to business-traveller behaviour, short lead times, and abundant city alternatives. Resort Hotels enjoy a stronger commitment profile (27.97% cancellation) due to long-term family vacation investments.

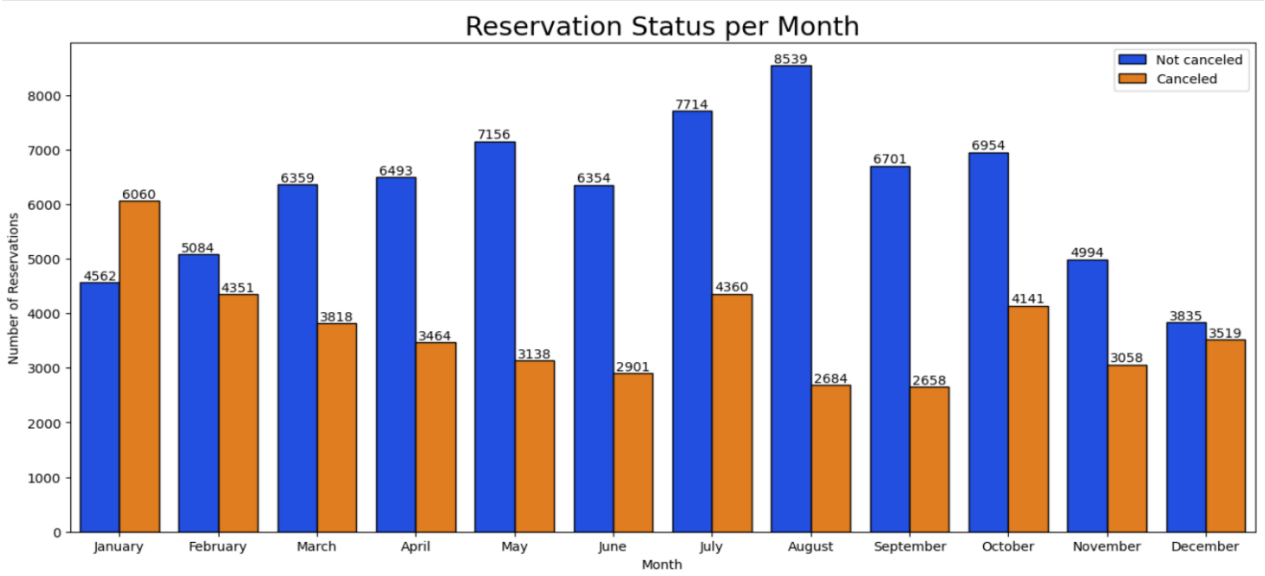


8. Monthly Reservation Analysis

An evaluation of monthly timelines establishes a highly seasonal distribution of demand and churn:

- **Peak Booking Volume: August (13,853 total bookings)** and **July (12,628 total bookings)** represent the high-season summer vacation peaks.

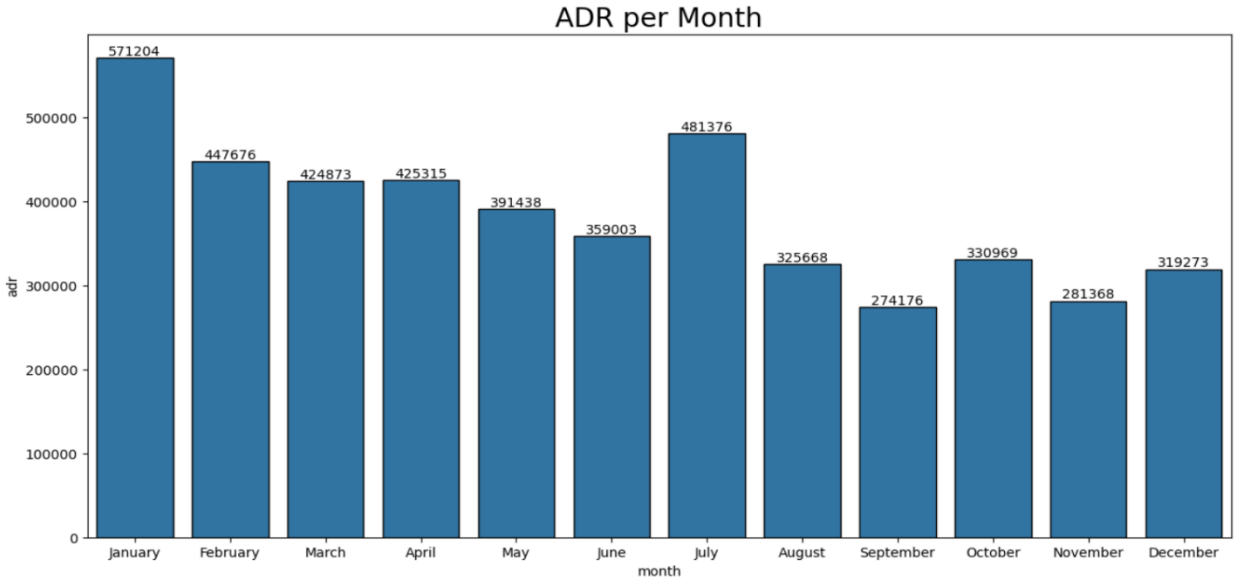
- **Trough Booking Volume:** January (5,874 bookings) and December (6,728 bookings) register the lowest demand profiles.
- **Cancellation Volumes:** Cancellations mirror the overall demand curve, peaking severely in August (5,237 cancellations) and July (4,736 cancellations), while the baseline cancellation percentage remains consistently high across the year.



9. ADR Analysis

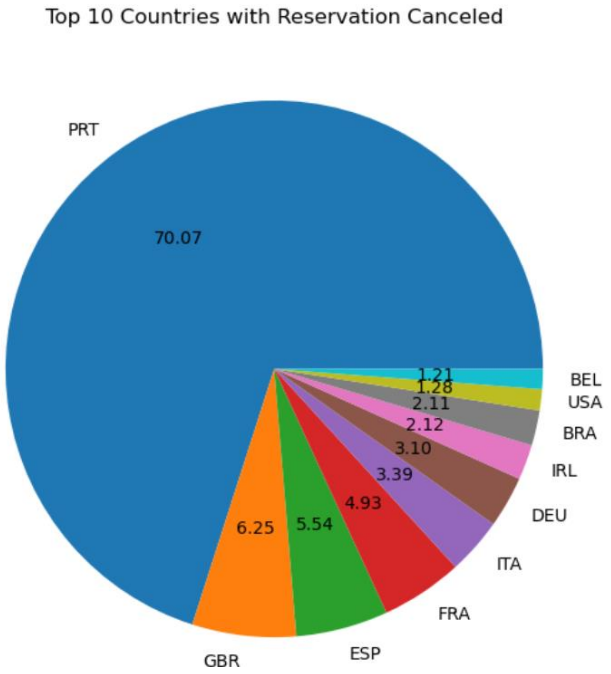
Analyzing chronological fluctuations of the Average Daily Rate (ADR) reveals completely different pricing strategies between the two hotel models:

- **Resort Hotels:** Exhibit extreme seasonal price elasticity. During peak summer periods (July–August), Resort ADR spikes aggressively, regularly surpassing City Hotel rates. In winter shoulder months, Resort ADR drops sharply to stimulate weak off-season demand.
- **City Hotels:** Maintain a highly stable, flat, and predictable ADR structure year-round. It experiences only minor increases during spring and autumn (corporate convention seasons), remaining largely unaffected by vacation calendar changes.



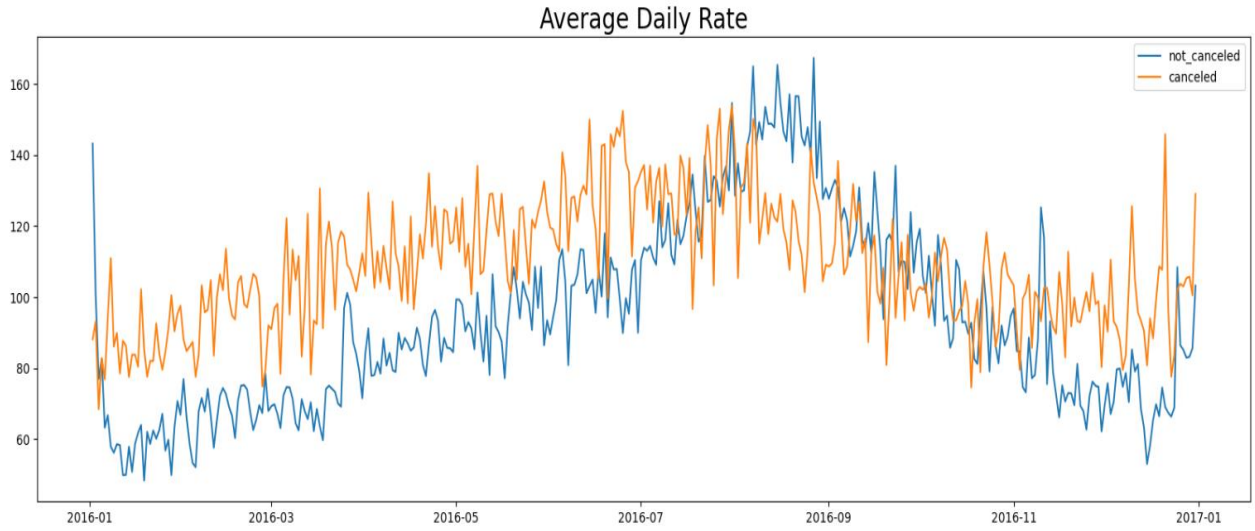
10. Country-wise Cancellation Analysis

Geographical mapping of cancellations reveals an extreme concentration of risk within the domestic market segment. When evaluating the top ten countries responsible for customer churn, **Portugal (PRT) accounts for an overwhelming 70.08%** of all canceled bookings. Great Britain (GBR) follows at a distant second with 5.67%, and Spain (ESP) at 4.70%. Local guests face minimal transit penalties and frequently use risk-free reservations as speculative safety nets.



11. ADR Comparison – Canceled vs Non-Canceled

A direct statistical comparison of ADR trends across the 2016–2017 timeline confirms a key operational theory: **canceled reservations consistently feature higher average daily rates than reservations that end in a check-out.** This indicates that high pricing directly induces cancellation behaviour. Guests booking at premium or uncalibrated high rates are highly susceptible to buyer's remorse or quickly switch to a competitor if a more economical alternative emerges.



12. Key Business Insights

- **Price-Induced Churn:** Higher average daily rates (ADR) are a primary catalyst for cancellations, making high-paying guests more volatile.
- **City Hotel Inventory Drain:** City hotels face an alarming 41.71% cancellation rate compared to 27.97% for resorts.
- **Domestic Market Risk:** Portuguese guests represent the vast majority of cancellations, showing that proximity decreases booking commitment.
- **Summer Peak Volatility:** July and August bring maximum booking volume but also the highest strain of absolute cancellations.
- **Segment Risk Concentration:** Online Travel Agents (Online TA) and group bookings comprise the overwhelming bulk of canceled slots.

13. Business Recommendations

- **Implement Non-Refundable Pricing:** For high-occupancy summer months (July-August) and high-ADR room selections, mandate upfront, non-refundable deposits or full pre-payment to lock in revenue.
- **Stricter Local Market Rules:** Introduce specialized, stricter booking terms or shorter cancellation windows specifically for domestic/Portuguese (PRT) market segments to prevent late-stage dropouts.
- **Predictive Overbooking:** Run a calculated overbooking strategy of 15% to 20% for City Hotels during peak months to counter the predictable 41.71% historical churn rate and avoid empty rooms.
- **Enforce Group Cut-Off Dates:** Enforce rigid contract terms for group blocks, requiring them to release unallocated rooms 30 days prior to arrival alongside mandatory deposits.
- **Recover Churned High-ADR Leads:** Automate email marketing campaigns to offer slightly discounted, completely non-refundable rates or value-added additions (e.g., spa credits) to recovered high-ADR guests immediately after they cancel.

14. Conclusion

Hotel cancellations follow distinct, predictable patterns related to property type, price sensitivity, seasons, and geography. Moving away from universal "free-cancellation" options toward targeted, data-backed policies—such as non-refundable premium tiers, tighter local market restrictions, structured group contracts, and calculated overbooking—will allow management to stabilize room occupancy, heavily minimize revenue leakages, and ensure sustainable business growth.